

RNE HW3-2 Report

An-Yi Lee

Student ID: S11259029

1 Introduction

This homework focuses on deep reinforcement learning in the Proly environment. Unlike HW3-1, the PPO algorithm, observation space, and action space are fixed, so the main task is to design a suitable reward function in `rl_play.py`. The goal is to guide the agent to capture checkpoints efficiently and complete the map more consistently. [1]

2 Implementation

2.1 Reward Design

I implemented the reward function in `rl_play.py` by combining several reward components.

First, I added a large positive reward when the value of `last_checkpoint_index` increases. This directly encourages the agent to capture a new checkpoint, which is the most important objective in this task.

Second, I added a distance-based reward using `target_position`. At each step, I compared the current distance to the target checkpoint with the previous one. If the agent moved closer to the target, it received a positive reward; otherwise, it received a negative reward. This dense reward helps the agent learn useful behavior before it is able to capture checkpoints consistently.

Third, I added a small survival reward and a penalty for health loss. This discourages dangerous behavior such as taking damage or moving into risky areas.

The final reward is the sum of checkpoint reward, distance reward, and survival-related reward.

2.2 Training Process

I trained the agent on the three public maps. At the beginning of training, the model was unstable and sometimes produced very low rewards because the agent could not consistently reach all checkpoints. After more training, the policy gradually improved and became more stable.

For map 1 and map 2, the trained agent was able to reach all 10 checkpoints consistently in later episodes. For map 3, the agent was initially less stable, so I continued training to improve robustness. After additional training, the agent also achieved stable full-checkpoint performance on map 3 in most later episodes.

3 Observation

The reward design worked well because it combines both sparse and dense signals. The checkpoint reward defines the main task objective, while the distance reward provides intermediate guidance. The survival and health penalty further reduce unstable or unsafe behavior.

From the training results, I observed that the agent gradually learned to move toward the target more efficiently and capture checkpoints with fewer unnecessary actions.

4 Conclusion

In this homework, I designed a reward function for the Proly task under a fixed PPO framework. By combining checkpoint reward, distance reward, and survival-related reward, the agent learned to complete the maps more consistently. This task helped me understand the importance of reward shaping in reinforcement learning, since the reward function directly affects the learning quality of the policy.

References

- [1] Robotic Navigation and Exploration, HW3: Deep Reinforcement Learning, course slides.